IPv6 Configuration and Subnetting Practical

Detailed step-by-step answers for Questions 1–5

Important: Replace example interface names and addresses with values suitable for your own network. Run commands only on a system you are allowed to configure. Record your real results; do not report a successful ping unless replies are received.

1. Assign a Static IPv6 Address and Verify It

Example address: fd12:3456:789a:1::10/64. Addresses beginning with fd are Unique Local Addresses (ULAs), intended for private networks. A static address alone does not guarantee internet access.

Windows: (1) Open Command Prompt and run `ipconfig` to record the adapter and current IPv6 settings. (2) Open Command Prompt as Administrator. (3) Find the exact adapter name:

```
netsh interface ipv6 show interfaces
```

(4) Add the address, replacing Wi-Fi with your actual interface name:

```
netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a:1::10
```

(5) Verify it:

```
ipconfig
netsh interface ipv6 show address
```

(6) Remove the test address after testing:

```
netsh interface ipv6 delete address "Wi-Fi" fd12:3456:789a:1::10
```

Linux: (1) Identify the interface using `ip link` or `ip -6 addr`. (2) Replace enp3s0 with your interface name:

```
sudo ip -6 addr add fd12:3456:789a:1::10/64 dev enp3s0
ip -6 addr show dev enp3s0
# Remove after testing:
sudo ip -6 addr del fd12:3456:789a:1::10/64 dev enp3s0
```

The Linux `ip addr add` command normally adds a runtime address that is not retained after reboot. On Windows, the manual netsh address above may persist until removed.

2. Configure a Unique Local IPv6 Address and Test LAN Connectivity

Use two different addresses from the same /64 subnet. Example:

Device	IPv6 address	Prefix
Computer A	fd12:3456:789a:1::10	/64
Computer B	fd12:3456:789a:1::20	/64

Steps: (1) Configure Computer A as fd12:3456:789a:1::10/64 and Computer B as fd12:3456:789a:1::20/64. Never assign the same address to both. (2) Connect both devices to the same LAN. (3) Ensure the firewall allows ICMPv6 echo requests. (4) From Computer A, run:

```
ping -6 fd12:3456:789a:1::20
```

On Windows use the command above; Linux also commonly supports it (some systems provide ping6). Replies confirm IPv6 reachability. A timeout may indicate an incorrect address/prefix, a disconnected device, separate networks, or a firewall blocking ping. Verify settings with ipconfig (Windows) or ip -6 addr (Linux).

3. Add a Temporary IPv6 Address and Document What Changes

This exercise uses a manually added runtime address. It is different from an OS-generated IPv6 privacy address, which is designed to reduce tracking.

Linux steps: (1) Create a text file named `ipv6_observations.txt`. (2) Record the OS, interface, date/time and initial output from `ip -6 addr show`. (3) Add the address:

```
sudo ip -6 addr add fd12:3456:789a:1::99/64 dev enp3s0
ip -6 addr show dev enp3s0
```

(4) Record whether the address appears. (5) Disconnect and reconnect the network, then check again. (6) Reboot and run `ip -6 addr show` once more. Record what actually happened.

A manually added Linux runtime address is normally lost after reboot and may be removed or affected by network-manager actions during reconnect. Behavior can vary by distribution and network settings. On Windows, a manually configured netsh address may remain after reboot until deleted; do not label it as a privacy-generated address.

Text-file template:

```
IPv6 temporary address test
OS/version:
Interface name:
Initial IPv6 addresses:
Command used:
Address after adding:
Address after disconnect/reconnect:
Address after reboot:
Conclusion: what changed and what stayed?
```

Fill the template with your own observations rather than invented results.

4. Restaurant Search Script Using IPv6 Addresses

This Python example stores restaurant names and sample IPv6 addresses, accepts a partial name, ignores capitalization, and prints every match. These are sample addresses for a programming exercise, not real Zomato service addresses.

```
restaurants = [
    {"name": "Spice Garden", "ipv6": "fd12:3456:789a:10::10"},
    {"name": "Urban Tandoor", "ipv6": "fd12:3456:789a:10::20"},
    {"name": "Green Bowl Cafe", "ipv6": "fd12:3456:789a:10::30"},
    {"name": "Spice Route", "ipv6": "fd12:3456:789a:10::40"},
]

search_name = input("Enter part of a restaurant name: ").strip().lower()
matches = [r for r in restaurants if search_name in r["name"].lower()]

if matches:
    for r in matches:
        print(r["name"], "->", r["ipv6"])
else:
    print("No matching restaurant found.")
```

How to run: (1) Save the code as `restaurant_ipv6_search.py`. (2) Run `python restaurant_ipv6_search.py` or `python3 restaurant_ipv6_search.py`. (3) Enter `spice`; Spice Garden and Spice Route should appear. (4) Try a name that is not in the array to test the no-match message.

5. IPv6 Subnetting Scenario and Manual Address Assignment

AI-generated scenario: An organization has the private IPv6 prefix fd12:3456:789a::/48. It will use three separate /64 networks: Office, Lab and Guest. Each subnet keeps the first three hextets and uses a different fourth hextet.

Purpose	Subnet prefix	Example host address
Office	fd12:3456:789a:1::/64	fd12:3456:789a:1::10/64
Lab	fd12:3456:789a:2::/64	fd12:3456:789a:2::10/64
Guest	fd12:3456:789a:3::/64	fd12:3456:789a:3::10/64

These are three distinct /64 subnets within the example /48. Choose one that matches your actual LAN. An address from a sample subnet will not automatically reach other devices unless those devices/network are configured consistently.

Linux configuration example: (1) Identify the interface using `ip link`. (2) Add the Lab host address, replacing enp3s0 with your interface:

```
sudo ip -6 addr add fd12:3456:789a:2::10/64 dev enp3s0
ip -6 addr show dev enp3s0
ping -6 fd12:3456:789a:2::20
```

The ping target is only an example peer; it replies only if another reachable device has that address and permits ICMPv6. Remove the test address when finished:

```
sudo ip -6 addr del fd12:3456:789a:2::10/64 dev enp3s0
```

Windows configuration example: (1) Open Command Prompt as Administrator. (2) Find the adapter name with `netsh interface ipv6 show interfaces`. (3) Add and verify:

```
netsh interface ipv6 add address "Ethernet" fd12:3456:789a:2::10
ipconfig
```

Replace Ethernet with the exact adapter name. Remove the address after testing:

```
netsh interface ipv6 delete address "Ethernet" fd12:3456:789a:2::10
```

Text-file submission: Create `ipv6_subnetting_steps.txt` and include (1) the scenario and three-subnet table, (2) OS and interface name, (3) exact command used, (4) actual verification output, and (5) ping result or a note explaining why it did not complete.

Quick Troubleshooting Notes

• **Address already exists:** Inspect the adapter's current addresses before adding another.

• **Ping fails:** Check the target, prefix, same-LAN connection, interface status and firewall ICMPv6 rules.

• **Address disappears:** Identify whether it was a runtime address, a privacy address, or a persistent network configuration.

• **No internet after setting a ULA:** An fd address is for private/local addressing; it does not itself provide internet routing. Preserve the normal default route and router configuration.